

Analysis 2: Midterm 1

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Problem 1.

Consider the function $g(t) = f(t)e^{-t}$ now doing MVT on this assuming a positive t we have,

$$\begin{aligned}g(t) &= g(0) + tg'(c) \quad \text{for some } c \text{ between } 0 \text{ and } t \\f(t)e^{-t} &= f(0)e^0 + t(f'(c)e^{-c} - f(c)e^{-c}) \\&= f(0) + te^{-c}(f'(c) - f(c)) \\f(t) &= f(0)e^t + te^{t-c}(f'(c) - f(c)) \\f(t) - f(0)e^t &= te^{t-c}(f'(c) - f(c))\end{aligned}$$

Now note that clearly $t \geq 0$ and clearly e^{t-c} is positive as well since e^x always returns a positive number. So this gives us,

$$te^{t-c} \geq 0$$

But now note that we have $f'(t) \leq f(t)$ for all t which means that $f'(c) - f(c) \leq 0$ so clearly we have,

$$(te^{t-c})(f'(c) - f(c)) \leq 0$$

Putting this back in we have,

$$f(t) - f(0)e^t = te^{t-c}(f'(c) - f(c)) \leq 0$$

Or that,

$$\begin{aligned}f(t) - f(0)e^t &\leq 0 \\f(t) &\leq f(0)e^t\end{aligned}$$

for any choice of t .

Problem 2.

We have $F : U \rightarrow \mathbb{R}^2$ a smooth vector field where $U \subseteq \mathbb{R}^2$. We also have $F = \nabla \times F = 0$. So if $F = (F_1, F_2)$ then we have $\partial_x F_2 - \partial_y F_1 = 0$ or $\partial_x F_2 = \partial_y F_1$

1. We have $U = B(0, 2)$. We need to find some $f : U \rightarrow \mathbb{R}$ such that $\nabla f = F$. Or we need some f such that $\partial_x f = F_1$ and $\partial_y f = F_2$ and note we have $\partial_x F_2 = \partial_y F_1$.

Now let r be some path from a point a to (x, y) . Now let p be another arbitrary path from the same starting and endpoints. Note now that if we move through r and then move through $-p$ we have a closed loop. We now define f as follows,

$$f(x, y) = \int_r F_1 dx + F_2 dy$$

Now note that if we consider $\int_r F_1 dx + F_2 dy - \int_p F_1 dx + F_2 dy$ then this forms a closed loop and because of greens theorem this is equal to zero. Point being from some fixed point a if we choose any path to (x, y) we have the same value for f . So we can choose some fixed point a and define,

$$f(x, y) = \int_a^{(x,y)} F_1 dx + F_2 dy$$

Now we can simply check that $\partial_x f = \partial_x \int_a^{(x,y)} F_1 dx = F_1$ and similarly we have $\partial_y f = F_2$ which means $\nabla f = F$.

2. If we have $U = B(0, 2) \setminus \{0\}$ we can't find a potential function for f . This is because when we remove the 0, the ball is no longer simply connected. So for any loop around the circle containing 0 we can never shrink it down to a single point.

Consider the function $F(x, y) = \left(-\frac{y}{x^2+y^2}, \frac{x}{x^2+y^2}\right)$. Clearly F is smooth in U because it is not defined at the singularities i.e $x, y = 0$. We can note that the curl is,

$$\partial_x F_2 - \partial_y F_1 = \frac{y^2 - x^2}{(x^2 + y^2)^2} - \frac{y^2 - x^2}{(x^2 + y^2)^2} = 0$$

However now consider a closed curve around 0, we know that the integral of F around this curve must be zero according to greens theorem. So let $r(t) = (\cos t, \sin t), 0 \leq t \leq 2\pi$, so we have $F(r(t))r'(t) = (-\sin t, \cos t)(-\sin t, \cos t) = \sin^2 t + \cos^2 t = 1$. So around the circle we have $\int_0^{2\pi} 1 dt = 2\pi \neq 0$. So we can never find a function f to satisfy this.

Problem 3.

We have the function $f(x, y) = x^4 + y^4$ given the constraint $x^2 + y^2 = 1$. We will use the notion of the lagrange multipliers to find the extreme values subject to the constraints. Let us define our constraint as the level set $G(x, y) = x^2 + y^2 = 1$. So we need, $\nabla f = \lambda \nabla G$.

We have $\nabla f = (4x^3, 4y^3)$ and we have $\nabla G = (2x, 2y)$. So we have the following equalities,

$$\begin{aligned} 4x^3 &= \lambda 2x \\ 4y^3 &= \lambda 2y \\ x^2 + y^2 &= 1 \end{aligned}$$

Solving the first two we get $2x^2 = \lambda$ OR $x = 0$. Clearly the second gives us the same i.e $2y^2 = \lambda$ or that $y = 0$. Now let us look at these cases,

Case 1. $x, y = 0$. Clearly $x^2 + y^2 = 0 \neq 1$. Not possible

Case 2. $2x^2 = \lambda, y = 0$. Here we have $x^2 + y^2 = 1$ so $x^2 = 1$ implies that $x = \pm 1$. In both cases we have $\lambda = 2$. So this case gives us two choices $(\pm 1, 0)$.

Case 3. $2y^2 = \lambda, x = 0$. x, y are symmetric so similar to case 2 we have, the choices $(0, \pm 1)$.

Case 4. $2x^2 = \lambda = 2y^2$. This gives us $x^2 = y^2$, so clearly we have $x^2 = y^2 = \frac{1}{2}$ so $x = y = \pm \frac{1}{\sqrt{2}}$. So we have 4 choices here, represented by,

$$\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}} \right)$$

So note that putting all our cases together we have 8 critical points represented by,

$$(\pm 1, 0), (0, \pm 1), \left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}} \right)$$

Now note that we're doing x^4 and y^4 in our equation of f so essentially we have an even function. Hence note we have,

$$f(\pm 1, 0) = 1, f(0, \pm 1) = 1, f\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

So the maximum value is 1 occurring at $(\pm 1, 0), (0, \pm 1)$ the minimum is $\frac{1}{2}$ at $\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}} \right)$